Probability and Statistics

Statistical intervals based on a single sample

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1 Basic Properties of Confidence Intervals

2 Large-Sample Confidence Intervals for a Population Mean and Proportion

3 Intervals Based on a Normal Population Distribution

Basic Properties of Confidence Intervals

- Confidence interval (置信区间)
- In Chapter 6, we have studied point estimator for a population parameter
- But because the point estimate obtained from a point estimator is a single number, hence by itself it provides no information about the precision and reliability of estimation
- A confidence interval is a range of values, derived from sample data, that is likely to contain the true population parameter with a specified level of confidence
- In other words, it provides an estimated interval around a sample statistic that, under repeated sampling, would capture the actual population parameter a certain percentage of the time (i.e., 95% of the time)

Basic Properties of Confidence Intervals

- If we have a random sample $X_1, X_2, \ldots, X_n$, with $\overline{X} = 1/n \sum_i X_i$ from a normal distribution with population mean μ and population standard deviation σ. Then, we have

$$P\left(-1.96 < \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} < 1.96\right) = .95$$

1. Multiply through by $\sigma/\sqrt{n}$:

$$-1.96 \cdot \frac{\sigma}{\sqrt{n}} < \overline{X} - \mu < 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

2. Subtract $\overline{X}$ from each term:

$$-\overline{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < -\mu < -\overline{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

3. Multiply through by -1 to eliminate the minus sign in front of μ (which reverses the direction of each inequality):

$$\overline{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}} > \mu > \overline{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

that is,

$$\overline{X} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$$

The equivalence of each set of inequalities to the original set implies that

$$P\left(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \overline{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) = .95$$

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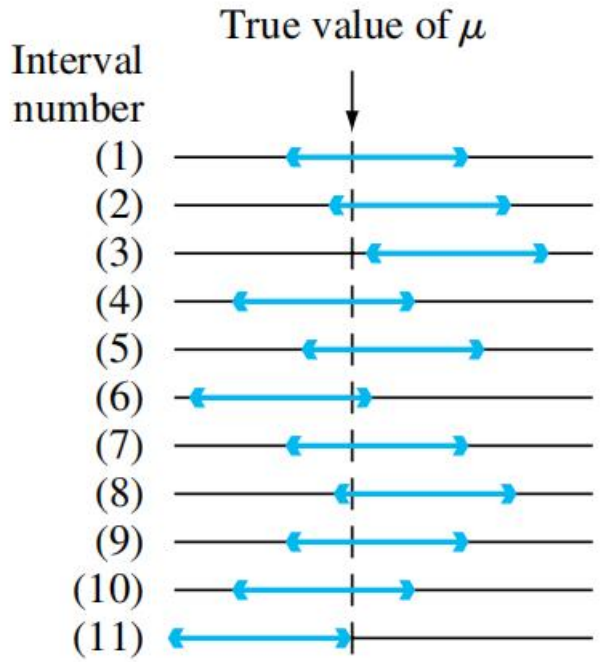
- 95% confidence interval for μ
- Or we can say that for any sample mean $\overline{X}$, with confidence level of 95%, the population mean μ is in

$$(\overline{X} - 1.96\frac{\sigma}{\sqrt{n}}, \overline{X} + 1.96\frac{\sigma}{\sqrt{n}})$$

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Basic Properties of Confidence Intervals

- A correct interpretation of "95% confidence" is the long-run relative frequency
- Recall that to say that an event A has probability 0.95 is to say that if the experiment on which A is defined is performed over and over again, in the long run, A will occur 95% of the time
- Let A be the event that $\overline{X} - 1.96\sigma/\sqrt{n} < \mu < \overline{X} + 1.96\sigma/\sqrt{n}$, since P(A) = 0.95, in the long run, 95% of our computed confidence intervals will contain μ



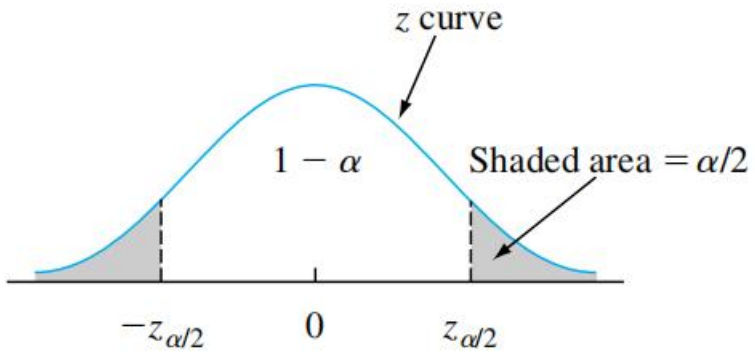
Basic Properties of Confidence Intervals

- Some definitions

A **100(1 − α)% confidence interval** for the mean μ of a normal population when the value of σ is known is given by

$$\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)$$

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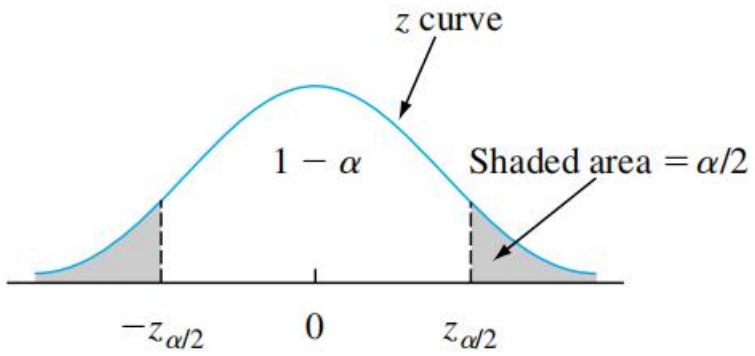
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or, equivalently, by $\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$.

z critical value (z临界值)

$$\Phi(z_{\alpha/2}) = 1 - \frac{\alpha}{2}$$

$$\Phi(-z_{\alpha/2}) = \frac{\alpha}{2}$$



Basic Properties of Confidence Intervals

- The relationship between confidence level and sample size
- Suppose the population probability distribution is a normal distribution with standard deviation σ, then how many sample data is needed to ensure that the resulting $100 * (1 - \alpha)\%$ confidence interval has a width of w?

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Basic Properties of Confidence Intervals

- The relationship between confidence level and sample size
- Suppose the population probability distribution is a normal distribution with standard deviation σ, then how many sample data is needed to ensure that the resulting $100 * (1 - \alpha)\%$ confidence interval has a width of w?
- Since the confidence interval is $\left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right)$

- The sample size n must satisfy $2 * z_{\alpha/2} * \sigma/\sqrt{n} = w$, so

$$n = \left(2z_{\alpha/2} \cdot \frac{\sigma}{w}\right)^2$$

Basic Properties of Confidence Intervals

- The relationship between confidence level and sample size
- Suppose the population probability distribution is a normal distribution with standard deviation of 25, then how many sample data is needed to ensure that the resulting 95% confidence interval has a width of 10?
- The sample size n must satisfy $2 * 1.96 * 25/\sqrt{n} = 10$, so n = 97

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Large-Sample Confidence Intervals

- Previously we assume that the population distribution is normal and the value of population standard deviation is known. If the sample size is large, we do not need those assumptions due to the central limit theorem

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- By the central limit theorem, when the sample size n is large, the sample mean $\overline{X} = 1/n \sum_i X_i$ is a normal random variable with mean of μ and standard deviation of $\sigma/\sqrt{n}$, so we have

$$P\left(-z_{\alpha/2} < \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} < z_{\alpha/2}\right) \approx 1 - \alpha$$ which means $100 * (1 - \alpha)\%$ large-sample confidence interval is

$$\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}.$$

- If the population standard devation σ is unknown, we can replace σ by sample standard deviation S in the sense that with large sample size sample standard deviation is very close to population standard deviation

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If n is sufficiently large, the standardized variable

$$Z = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

has approximately a standard normal distribution. This implies that

$$\bar{x} \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

is a **large-sample confidence interval for μ** with confidence level approximately $100(1 - \alpha)\%$. This formula is valid regardless of the shape of the population distribution.

Large-Sample Confidence Intervals

62	50	53	57	41	53	55	61	59	64	50	53	64	62	50	68
54	55	57	50	55	50	56	55	46	55	53	54	52	47	47	55
57	48	63	57	57	55	53	59	53	52	50	55	60	50	56	58

Summary quantities include $n = 48$, $\sum x_i = 2626$, and $\sum x_i^2 = 144{,}950$, from which $\bar{x} = 54.7$ and $s = 5.23$. The 95% confidence interval is then

$$54.7 \pm 1.96\frac{5.23}{\sqrt{48}} = 54.7 \pm 1.5 = (53.2, 56.2)$$

That is,

$$53.2 < \mu < 56.2$$

with a confidence level of approximately 95%. The interval is reasonably narrow, indicating that we have precisely estimated μ.

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Large-Sample Confidence Intervals

- Confidence interval for a population proportion
- Let p denote the proportion of "successes" in a population, where success identifies an individual or object that has a specified property, we want to obtain confidence interval of p

Large-Sample Confidence Intervals

- Confidence interval for a population proportion
- Let p denote the proportion of "successes" in a population, where success identifies an individual or object that has a specified property, we want to obtain confidence interval of p
- We first select a random sample of n individuals, and X is the number of successes in the sample, then since X is binomial random variable with $E(X) = np$ and $\sigma_X = \sqrt{np(1-p)}$, but at the same time, since X is a sum of random variables, it is approximately a normal random variable (if $np \geq 10$ and $n(1-p) \geq 10$)
- If we define the estimator of p as $\hat{p} = X/n$, $\hat{p}$ is also a normal random variable with $E(\hat{p}) = p$ and $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$ such that we have

$$P\left(-z_{\alpha/2} < \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} < z_{\alpha/2}\right) \approx 1 - \alpha$$

A **confidence interval for a population proportion** p with confidence level approximately $100(1-\alpha)\%$ has

$$\text{lower confidence limit} = \frac{\hat{p} + \frac{z^2_{\alpha/2}}{2n} - z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z^2_{\alpha/2}}{4n^2}}}{1 + (z^2_{\alpha/2})/n}$$

and

$$\text{upper confidence limit} = \frac{\hat{p} + \frac{z^2_{\alpha/2}}{2n} + z_{\alpha/2}\sqrt{\frac{\hat{p}\hat{q}}{n} + \frac{z^2_{\alpha/2}}{4n^2}}}{1 + (z^2_{\alpha/2})/n}$$

If the sample size is quite large, $z^2/(2n)$ is negligible compared to $\hat{p}$, $z^2/(4n^2)$ under the square root is negligible compared to $\hat{p}\hat{q}/n$, and z^2/n is negligible compared to 1. Discarding these negligible terms gives approximate confidence limits

$$\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$$

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Large-Sample Confidence Intervals

- If a worker is checking products, and suppose there are 16 defective products from 48 products. What is 95% confidence interval for p ?

Large-Sample Confidence Intervals

- If a worker is checking products, and suppose there are 16 defective products from 48 products. What is 95% confidence interval for p ?
- Since $\hat{p} = 16/48 = 0.33, \hat{q} = 1 - 16/48 = 0.667, z_{0.05/2} = 1.96$, the confidence interval of p with confidence level 95% is

A **confidence interval for a population proportion** p with confidence level approximately $100(1-\alpha)\%$ has

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$$\frac{.333 + (1.96)^2/96 \pm 1.96\sqrt{(.333)(.667)/48 + (1.96)^2/9216}}{1 + (1.96)^2/48}$$

$$= \frac{.373 \pm .139}{1.08} = (.217, .474)$$

If the sample size is quite large, $z^2/(2n)$ is negligible compared to $\hat{p}$, $z^2/(4n^2)$ under the square root is negligible compared to $\hat{p}\hat{q}/n$, and z^2/n is negligible compared to 1. Discarding these negligible terms gives approximate confidence limits

$$\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$$



$$.333 \pm 1.96\sqrt{(.333)(.667)/48} = .333 \pm .133 = (.200, .466)$$

Large-Sample Confidence Intervals

- One-Sided confidence intervals
- The confidence intervals discussed thus far give both a lower confidence bound and an upper confidence bound for the parameter being estimated
- In some circumstances, an investigator will want only one of these two types of bounds

Because the cumulative area under the standard normal curve to the left of 1.645 is .95,

$$P\left(\frac{\bar{X}-\mu}{S/\sqrt{n}} < 1.645\right) \approx .95$$



$$\mu > \bar{x} - 1.645s/\sqrt{n};$$

Large-Sample Confidence Intervals

- One-Sided confidence intervals
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A **large-sample upper confidence bound for μ** is

$$\mu < \bar{x} + z_\alpha \cdot \frac{s}{\sqrt{n}}$$

and a **large-sample lower confidence bound for μ** is

$$\mu > \bar{x} - z_\alpha \cdot \frac{s}{\sqrt{n}}$$

A **one-sided confidence bound for p** results from replacing $z_{\alpha/2}$ by z_α

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Intervals Based on a Normal Population Distribution

- The confidence interval constructed previously is valid if sample size n is large
- The key result is based on the fact that for large n, the random variable $(\bar{X} - \mu)/(S/\sqrt{n})$ has approximately a standard normal distribution
- But what if the sample size n is small?

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- But what if the sample size n is small?
- t probability distribution:

When $\bar{X}$ is the mean of a random sample of size n from a normal distribution with mean μ, the rv

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has a probability distribution called a t distribution with $n - 1$ degrees of freedom (df).

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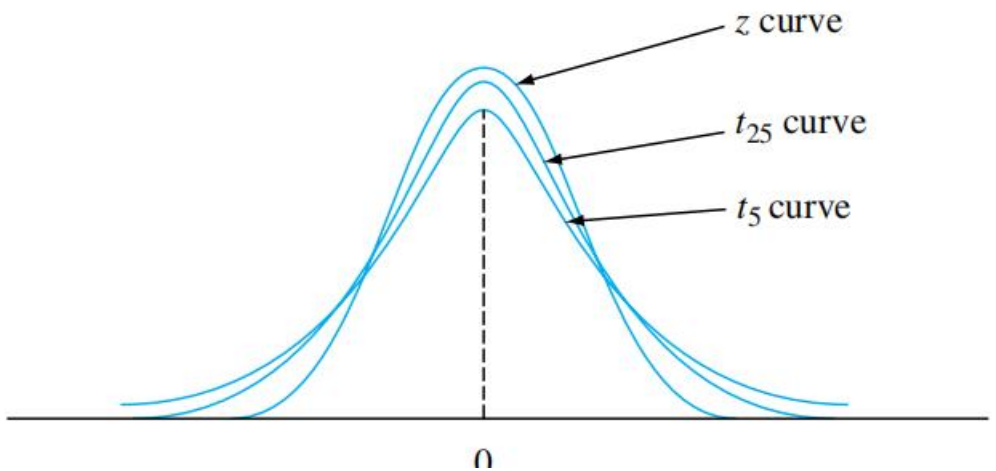
The degree of freedom (自由度） is usually dnoted by ν

Intervals Based on a Normal Population Distribution

- Properties of t distribution

Let t_ν denote the density function curve for ν df.

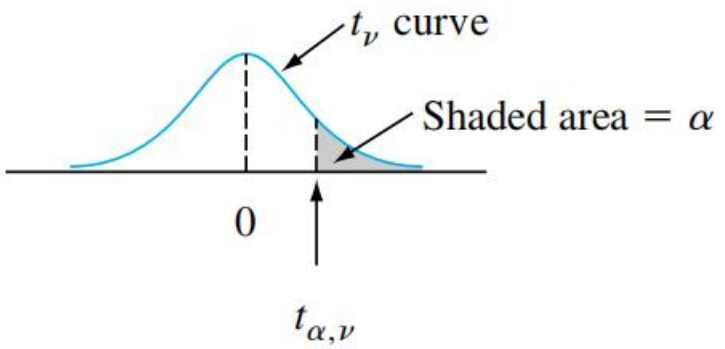
1. Each t_ν curve is bell-shaped and centered at 0.
2. Each t_ν curve is more spread out than the standard normal (z) curve.
3. As ν increases, the spread of the corresponding t_ν curve decreases.
4. As $\nu \longrightarrow \infty$, the sequence of t_ν curves approaches the standard normal curve (so the z curve is often called the t curve with df $= \infty$).



Intervals Based on a Normal Population Distribution

- Critical value of t distribution

Let $t_{\alpha,\nu}$ = the number on the measurement axis for which the area under the t curve with ν df to the right of $t_{\alpha,\nu}$ is α; $t_{\alpha,\nu}$ is called a ***t* critical value.**



Appendix Table A.5 gives $t_{\alpha,\nu}$

Intervals Based on a Normal Population Distribution

Let $\bar{x}$ and s be the sample mean and sample standard deviation computed from the results of a random sample from a normal population with mean μ. Then a **100(1 − α)% confidence interval for μ** is

$$\left(\bar{x} - t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}, \bar{x} + t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}\right)$$

or, more compactly, $\bar{x} \pm t_{\alpha/2,n-1} \cdot s/\sqrt{n}$.

An **upper confidence bound for μ** is

$$\bar{x} + t_{\alpha,n-1} \cdot \frac{s}{\sqrt{n}}$$

and replacing + by − in this latter expression gives a **lower confidence bound for μ,** both with confidence level $100(1 - \alpha)\%$.

Intervals Based on a Normal Population Distribution

- Suppose the sample size is 16, and sample mean is $\overline{X} = 14{,}532.5$, and sample variance is $S = 2055.67$. The sample size is 16, so the degree of freedom is 15.
- A confidence level of 95% for a two-sided interval requires the t critical value of $t_{0.05/2,14} = 2.131$. So the resulting interval is?

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$$\bar{x} \pm t_{.025,15} \cdot \frac{s}{\sqrt{n}} = 14{,}532.5 \pm (2.131)\frac{2055.67}{\sqrt{16}}$$

$$= 14{,}532.5 \pm 1095.2 = (13{,}437.3,\ 15{,}627.7)$$

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